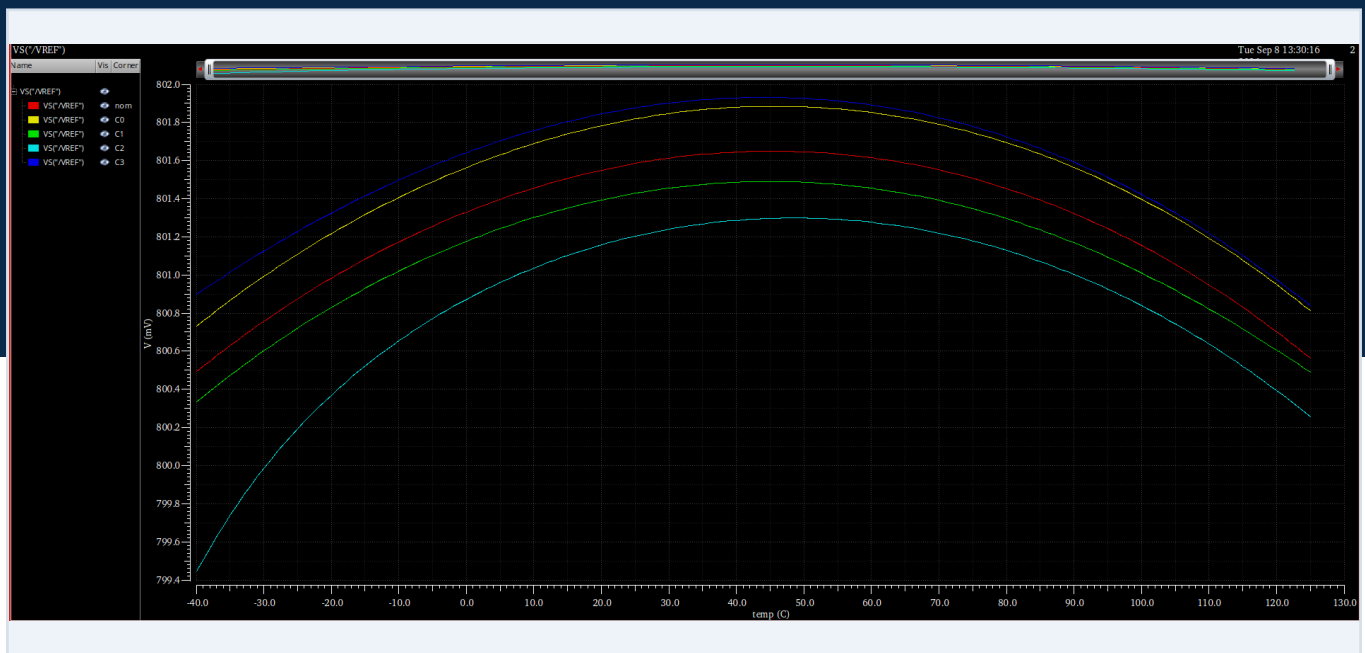


Self-biased sub-1 V bandgap reference

CTAT/PTAT synthesis, transistor-level loop design, and a comparison of ideal and PDK passives.

SCREENSHOT-BACKED RECONSTRUCTION



Selected simulation evidence. Full conditions and interpretation appear in the report.

REFERENCE TARGET

0.8 V

SUPPLY

1.2 V

TEMPERATURE

-40 to 125 C

Compiled from the author's original report, screenshots, and saved simulation records. Simulations were not rerun for this edition.

One reference. Three design stages.

The design generates a sub-1 V output from a 1.2 V supply by combining the negative temperature slope of a PNP base-emitter voltage with a scaled positive-temperature-coefficient term. A self-biased core is first closed with a behavioral amplifier, then a transistor-level OTA, and finally PDK resistor and capacitor cells.

Implementation	Nominal DC result	Temperature envelope
Behavioral amplifier	VREF ~806.7 mV; IDD 5.908 uA	~805.08 to 807.07 mV
Transistor OTA / ideal passives	VREF ~801.6 mV; IDD 6.303 uA	~799.44 to 801.94 mV
Transistor OTA / PDK passives	VREF ~803.6 mV; IDD 9.162 uA	~800.34 to 803.59 mV

Nominal values come from annotated operating points. Temperature envelopes are approximate readings of the archived plots. They are distinct implementations and must not be merged into one result set.

DESIGN BRIEF

Requirement	Assignment baseline
Process and supply	65 nm; VDD = 1.2 V
Output reference	Approximately 0.8 V
Temperature sweep	-40 to 125 degrees C
MOS process corners	TT, FF, SS, FS, SF
Total bias current	Less than 10 uA
Phase margin	Greater than 60 degrees

Verification scope

The archived corner setup varies MOS models while BJT and passive models remain typical. The supply-current readings are nominal operating points. This evidence does not establish full-PDK PVT or current-corner compliance.

Balancing CTAT and PTAT.

The PNP base-emitter voltage decreases with temperature. Two junctions operated at different current densities create a positive-temperature-coefficient difference voltage. Resistor scaling combines the two terms into a reference near 0.8 V.

Term	Source-grounded value or relation
CTAT slope	$dV_{BE}/dT = -1.921 \text{ mV/K}$
Emitter/current-density ratio	$n = 24$
Thermal voltage at 27 C	$V_T \sim 25.88 \text{ mV}$
PTAT voltage	$\Delta V_{BE} = V_T \ln(n) \sim 82.24 \text{ mV}$
PTAT current	$I_{PTAT} = \Delta V_{BE} / R_2$



Figure 1. Extracted PNP VBE versus temperature; the companion derivative screenshot reports -1.921 mV/K.

Corrected hand-calculation note

The original report associates 82.24 mV / 700 nA with 82.24 kohm. The quotient is 117.5 kohm; 82.24 kohm instead corresponds to 1 uA. The later simulation-tuned R2 value is 91.4 kohm and is documented independently.

The final resistor values are determined by iterative temperature-slope and nominal-output tuning, so the initial algebra and the selected simulation values should be read as separate design steps.

From first-pass sizing to the core.

Device group	Bias / gm/ID	L / W
Core PMOS mirror	2 uA / 10 per V	3.96 um / 10.48 um
OTA NMOS input pair	1.5 uA / 25 per V	2.00 um / 82.56 um
OTA NMOS mirror	1 uA / 20 per V	3.82 um / 14.8 um

These are documented sizing selections from the source report, not a consolidated extracted netlist. They establish the gm/ID-based starting point for the regulation loop.

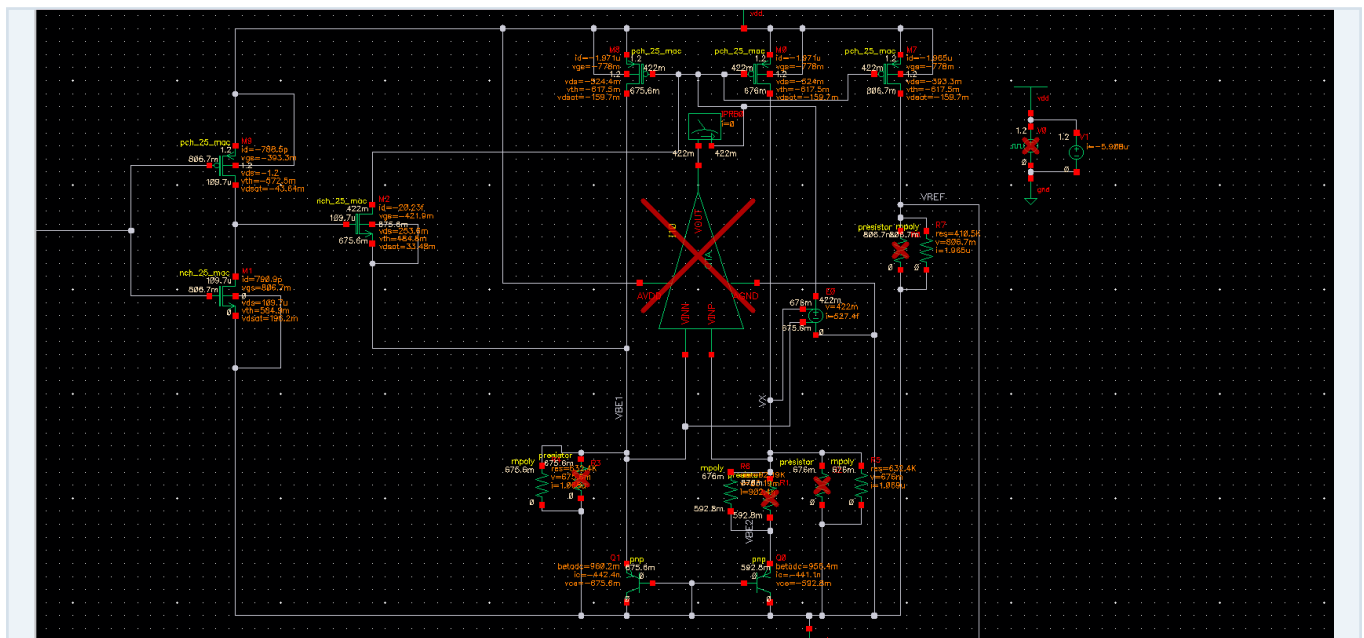


Figure 2. Behavioral-amplifier implementation and annotated DC operating point.

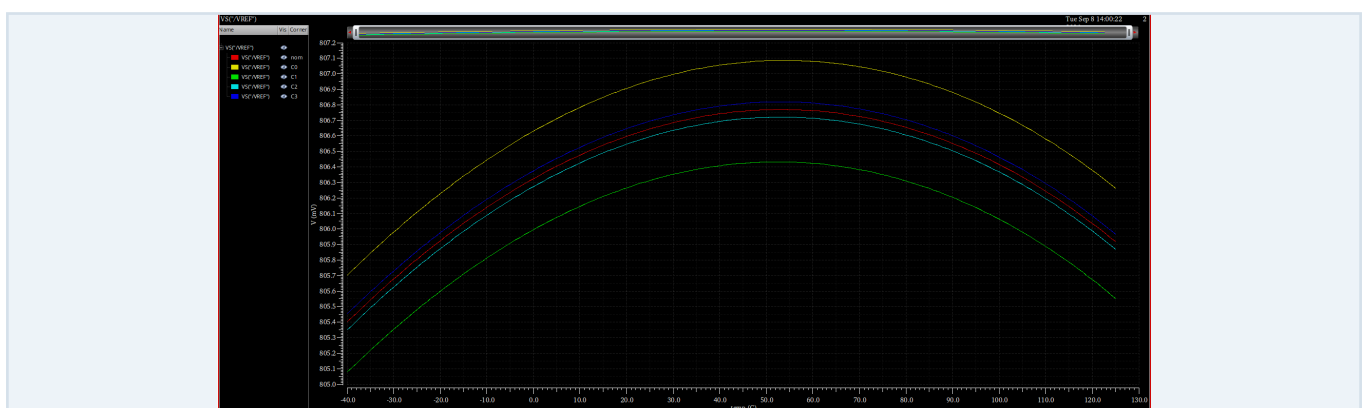


Figure 3. Behavioral temperature sweep, approximately 805.08 to 807.07 mV across the displayed family.

Transistor OTA with ideal passives.

Replacing the behavioral amplifier closes the self-biased reference with the designed OTA. Resistor sweeps tune the PTAT/CTAT balance and output level. The final saved variables are $R2 = 91.4 \text{ kohm}$, $R3 = 610 \text{ kohm}$, and $R4 = 400 \text{ kohm}$.

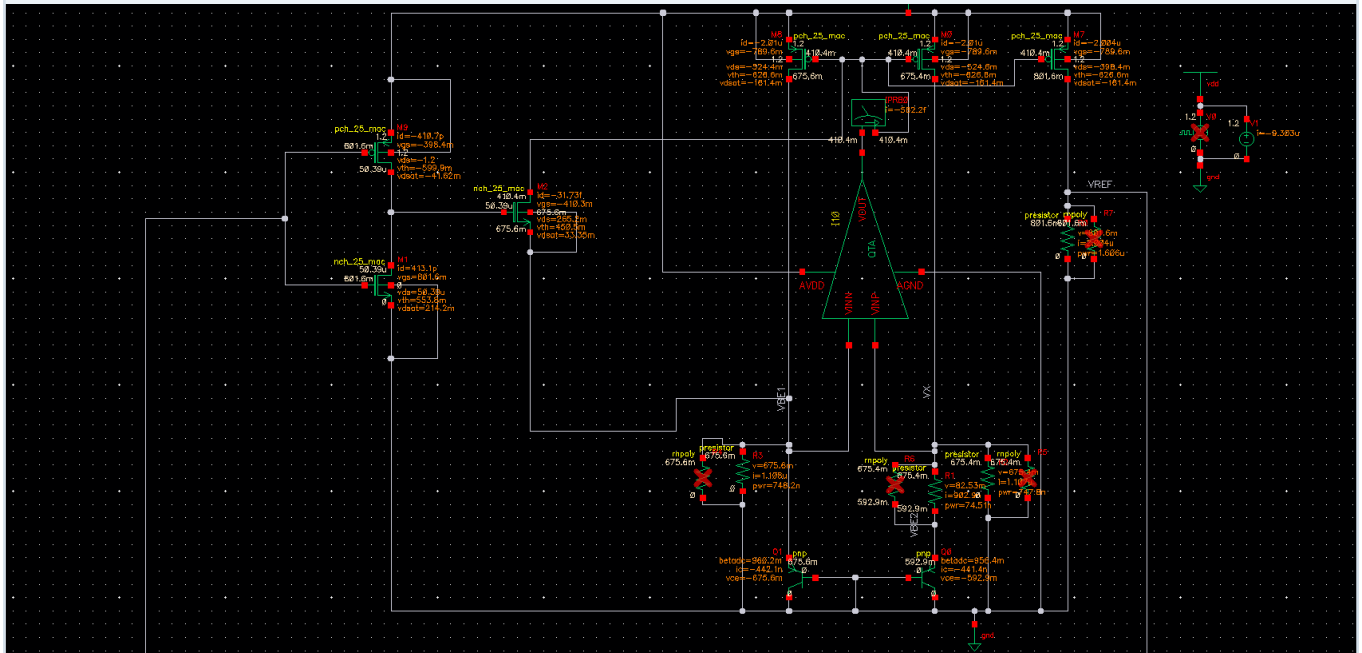


Figure 4. Final ideal-passive reference core. The annotated operating point shows $VREF \sim 801.6 \text{ mV}$ and nominal supply current 6.303 uA .

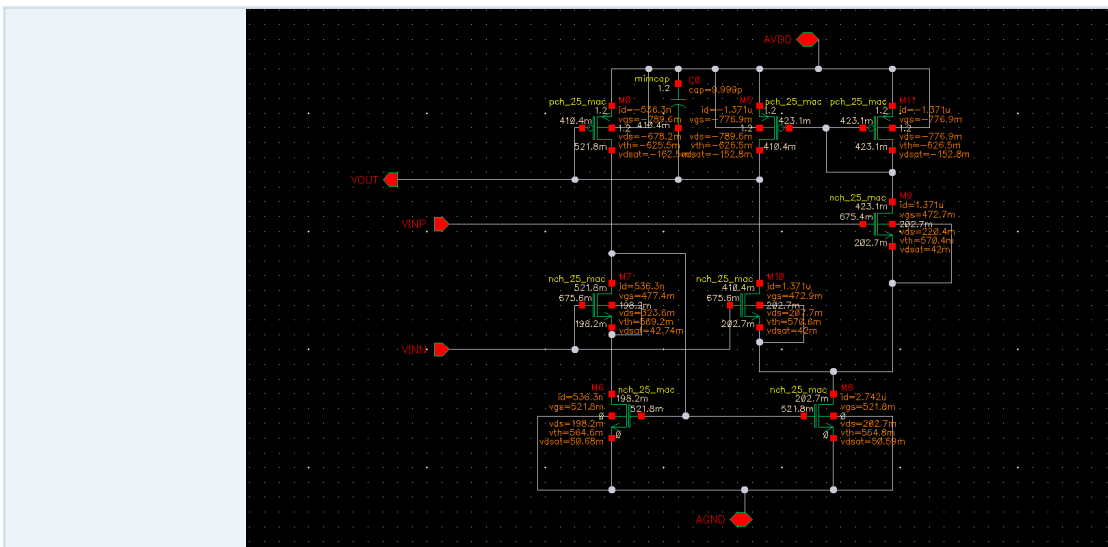


Figure 5. Transistor-level OTA operating point used in the reference loop.

The nominal current is 3.697 uA below the 10 uA brief limit. No current-versus-temperature/corner sweep is available in the source evidence.

Reading the temperature envelope.

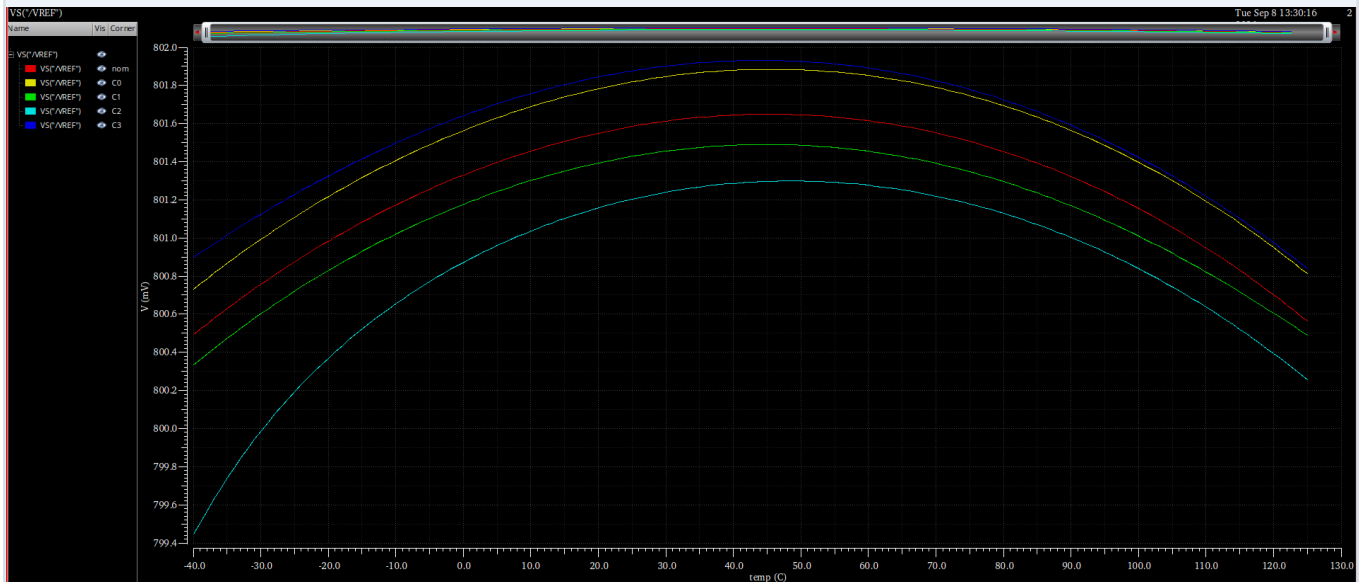


Figure 6. VREF over temperature for the evaluated MOS corner family with ideal passive elements.

Plot-derived metric	Approximate result
Lowest / highest visible VREF	799.44 mV / 801.94 mV
Global all-corner envelope	2.49 mV peak-to-peak
Worst single-corner temperature span	1.86 mV
Original report description	Delta around 2 mV

The original 'around 2 mV' statement is compatible with a within-corner interpretation. The all-corner envelope is closer to 2.5 mV. This report states the definition whenever a spread is quoted.

Corner coverage

The saved setup maps C0 = FF, C1 = SS, C2 = FS, and C3 = SF for MOS models. BJT remains tt_bip and the passive model selections remain typical. Temperature and MOS-corner coverage should not be described as a complete foundry PVT sign-off.

Compensation and loop stability.

A compensation capacitor of approximately 10 pF is used for the documented transistor-level loop. The captured loop-magnitude family and phase-margin sweep characterize the ideal-passive implementation.

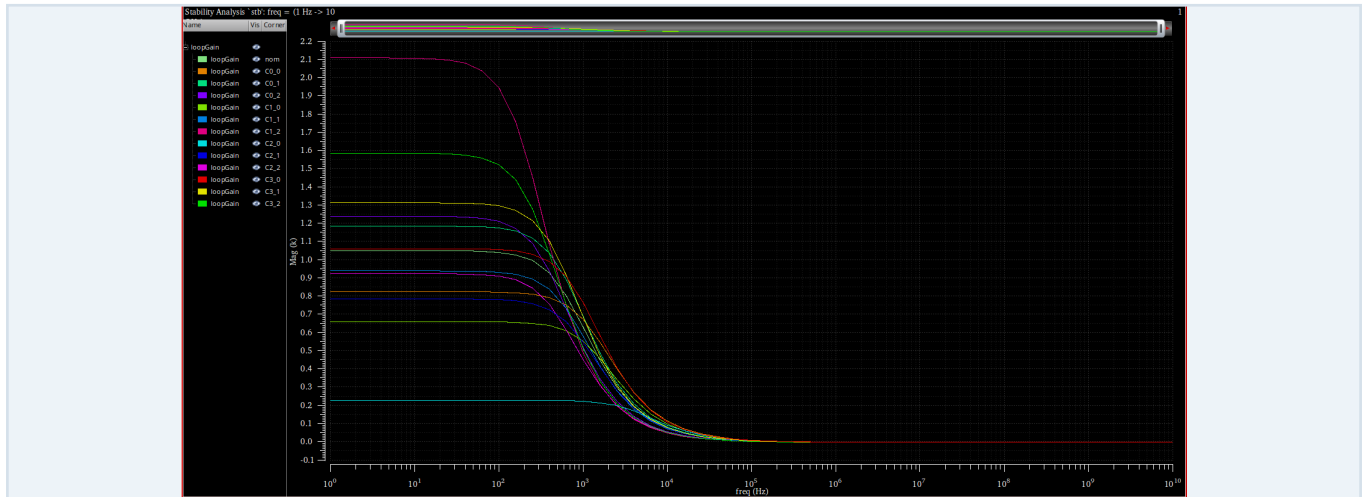


Figure 7. Loop-gain magnitude across the evaluated family. The vertical scale is linear magnitude in thousands, not decibels.



Figure 8. Phase margin versus temperature and MOS corner, approximately 65.7 to 78.5 degrees.

Stability result	Evidence
Minimum visible PM	~65.7 degrees; ~5.7 degrees above the brief
Maximum visible PM	~78.5 degrees
Applicability	Part 2 ideal-passive core; no equivalent Part 3 sweep saved

Supply-ramp startup behavior.

The saved transient test exercises the self-biased loop with a rising and falling supply ramp. The reference leaves the zero-output state and moves toward its regulated level after the supply rises.



Figure 9. Reference and supply response during the captured ramp cycle. The ramp interval is on the order of 1 ms.

WHAT THE CAPTURE ESTABLISHES

The waveform supports qualitative startup under the displayed ramp test. It also makes the transition behavior during supply rise and fall visible for later refinement. Repeated-startup reliability requires further trials.

WHAT REQUIRES AN ADDITIONAL MEASUREMENT

No saved cursor establishes a startup time, overshoot percentage, or settling-time limit. A robust verification should sweep supply-ramp rate, initial conditions, temperature, and process corners, and measure the time to a stated reference-error band.

Interpretation

Transient overshoot is a property of the complete nonlinear startup path and loop response. The screenshot alone does not establish a single causal link to phase margin.

Retuning with PDK resistor cells.

The final implementation replaces ideal passives with rnpoly resistors and a mimcap compensation capacitor. Their model behavior changes the temperature response and operating point, requiring a separate resistor retune.

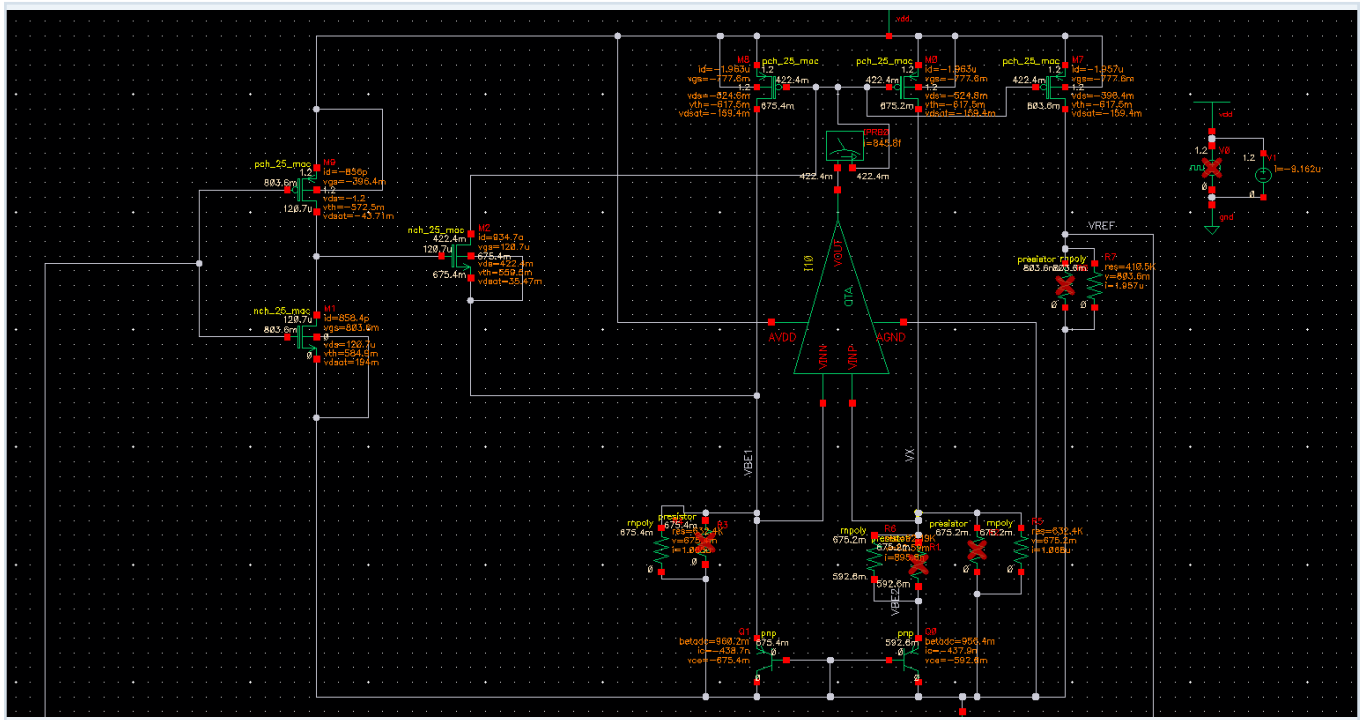


Figure 10. PDK-passive reference core. Nominal annotations show VREF ~803.6 mV and IDD = 9.162 uA.

Parameter	Part 2: ideal	Part 3: PDK
R2 design variable	91.4 kohm	91.4 kohm
R3 design variable	610 kohm	627 kohm
R4 design variable	400 kohm	407 kohm
Nominal VREF	~801.6 mV	~803.6 mV
Nominal IDD	6.303 uA	9.162 uA
Nominal current margin to 10 uA	3.697 uA	0.838 uA

The table lists design-variable values. PDK device annotations can show effective resistance values that differ from the variables. The smaller Part 3 current margin is nominal-only.

The PDK-passive result, separately.

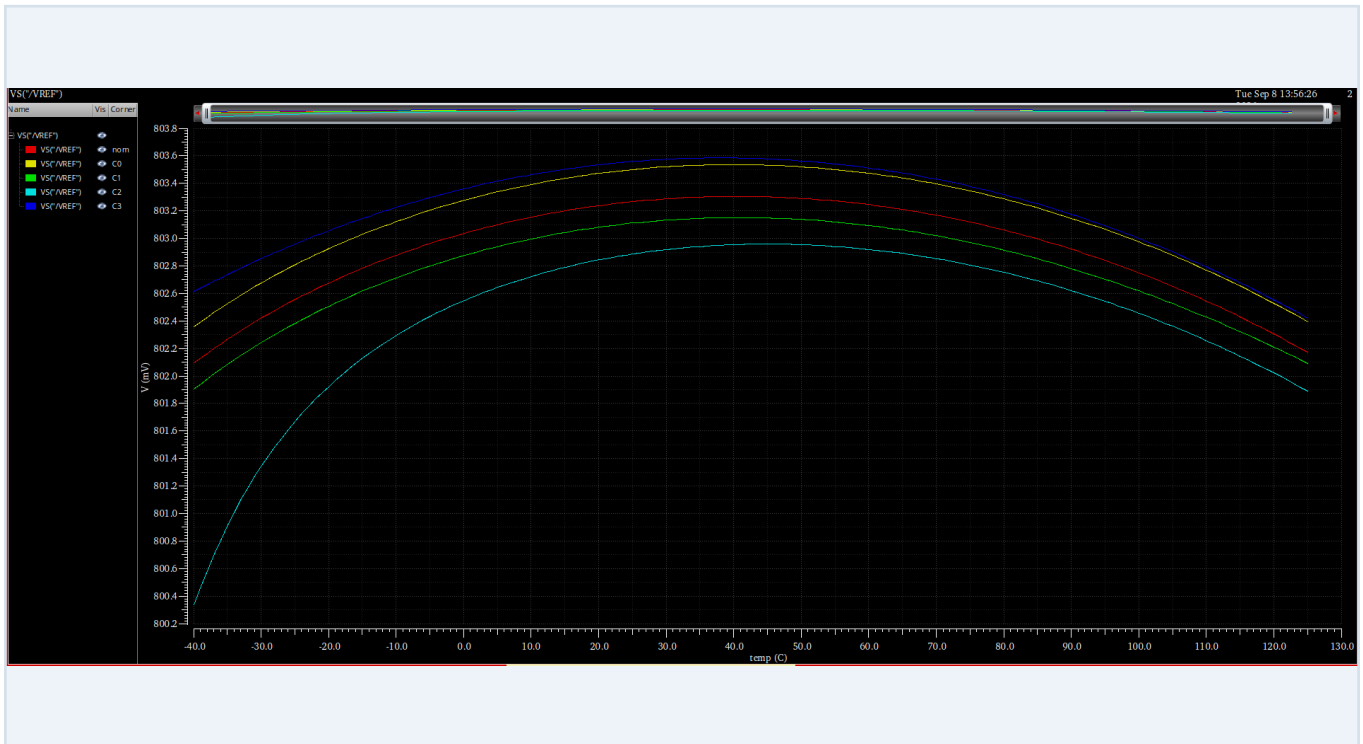


Figure 11. PDK-passive VREF family across the evaluated temperature and MOS-corner sweep.

Plot-derived metric	Approximate result
Lowest / highest visible VREF	800.34 mV / 803.59 mV
Global all-corner envelope	3.25 mV peak-to-peak
Worst single-corner temperature span	2.62 mV
Nominal offset from 800 mV	~+3.6 mV (+0.45%)
Compensation implementation	mimcap; annotated value ~9.999 pF

The final PDK-passive core remains close to the 0.8 V target, with a larger visible all-corner envelope than the ideal-passive version. The screenshot family is the evidence for this result; the earlier Part 2 phase-margin range must not be transferred to it.

Next verification priority

Repeat loop stability and startup after the passive replacement, then evaluate current and reference error over complete model and supply corners. The source archive contains no such Part 3 closure data.

What is demonstrated.

Objective	Evidence and status
Sub-1 V reference	Demonstrated near 0.8 V in all three saved implementations.
Low nominal supply current	6.303 μA with ideal passives; 9.162 μA with PDK passives. Both nominal readings are below 10 μA .
Temperature compensation	Visible -40 to 125 C sweeps; global envelope ~ 2.49 mV ideal and ~ 3.25 mV PDK.
Phase margin above 60 degrees	Documented for Part 2 ideal passives: ~ 65.7 to 78.5 degrees.
Self-biased startup	Qualitatively demonstrated by the Part 2 ramp transient.
PDK passive implementation	rnpoly / mimcap substitution and resistor retuning are documented.
Full-PDK PVT, mismatch, layout	Not established by the provided evidence.

The strongest result is a traceable progression from CTAT/PTAT theory to a transistor-level self-biased reference and then to PDK passive cells. Separating each stage reveals the effect of passive models on nominal current, reference level, and temperature spread.

RECOMMENDED FOLLOW-UP

Export raw waveforms and the selected netlist; define global versus within-corner temperature error; repeat STB and startup on the PDK-passive core; and measure supply current, line sensitivity, and startup metrics across the intended operating space.

Report provenance: the author's 19-page 'BGR Report.pdf', the two-page 'Self-Biased Sub-1V Bandgap Reference Circuit' brief, and the original September 8 screenshot archive. This edition reorganizes and source-checks those materials; it does not introduce new simulation runs.

Evidence index.

The following original captures are embedded or directly referenced. Paths are relative to the source archive's 'BGR design challenge 2026' folder. The assignment brief is used to define the design objectives, not as a simulation result.

SOURCE SCREENSHOTS

- 01. corners / Screenshot 2026-09-08 203045.png
- 02. extracted b / Screenshot 2026-09-08 174906.png
- 03. extracted b / Screenshot 2026-09-08 174916.png
- 04. behav / Screenshot 2026-09-08 210144.png
- 05. behav / Screenshot 2026-09-08 210240.png
- 06. DCop / Screenshot 2026-09-08 202641.png
- 07. DCop / Screenshot 2026-09-08 202654.png
- 08. corners / Screenshot 2026-09-08 203601.png
- 09. corners / for LG / Screenshot 2026-09-08 203910.png
- 10. corners / for LG / Screenshot 2026-09-08 203935.png
- 11. trans / Screenshot 2026-09-08 204550.png
- 12. real poly and mim / Screenshot 2026-09-08 205856.png
- 13. real poly and mim / Screenshot 2026-09-08 205643.png
- 14. real poly and mim / Screenshot 2026-09-08 205541.png

Source screenshots retain their original labels and measurement annotations. Approximate values in this report are derived from visible axes; original waveform data is required for greater precision.